- **Shared software:** Application servers (see page 86) can provide access to shared software. Some servers can provide access to operating systems.
- **Shared storage and file access:** The amount of storage available to users can be managed centrally. Sharing storage means that users can make files available to others. File permissions can be set for individual files, folders or drives, allowing users to either read only or read and write to different files.
- **Roaming profiles:** This is the ability to log into any computer in an office and see your settings and files. This allows users to access data, applications, mail and printers from any client, enabling them to work from anywhere where there is a client.

HINT

You learned about sharing software, storage and file access in *Unit 1 Digital devices* (pages 4–63).

SECURING DATA ON NETWORKS

Security prevents unauthorised users from accessing network resources and data.

LOGINS AND PASSWORDS

Users log in to computers on a network to access centrally managed resources. Without the correct **login** details, users cannot access the network or its resources.

Passwords are used to **authenticate** a user to the network. An authentication server can be used to manage the authentication of a user to a range of network resource and services. This means that, once a user has been authenticated, they do not have to log in each time they access a resource or service.

See page 86 for more information about authentication servers. You will learn more about passwords in *Unit 3 Operating online* (page 100).

SUBJECT VOCABULARY

login user or account information, such as a user name or account name
authenticate confirm that the user is who they say they are

FIREWALLS

A firewall is used at the gateway to a network. It controls the network traffic to and from a network, particularly the traffic from the internet. Firewalls prevent unauthorised users from accessing network devices and resources, such as storage. Firewalls are available as hardware and software, which can be installed on computers to prevent attacks from within a network.

ENCRYPTION

Encryption is the process of encoding, scrambling or jumbling data so that unauthorised users are prevented from being able to understand it.

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
CREATIVITY
ADAPTABILITY
SELF-DIRECTION
COMMUNICATION

ACTIVITY

▼ ANCIENT ENCRYPTION

Research how a tool called a scytale was used by Spartans to encrypt messages. Create your own scytale and then use it to produce an encrypted message.

SUBJECT VOCABULARY

cipher a code

One method used to encrypt text is called a Caesar **cipher**. This method shifts each letter to the left by a set number of places. The number of places by which the letters have been shifted is known as the key.

Figure 5.18 shows an example of using a Caesar cipher. The top row is the original alphabet. The bottom row shows the alphabet after applying a Caesar cipher with a shift of four places. The example shows how the original letter F is encrypted to B.



▲ **Figure 5.18** Once you know the key, it is easy to decrypt the message

SKILLS PROBLEM SOLVING
INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ DECRYPTING A MESSAGE

Using Figure 5.18, decrypt this message: WHWJ PQNEJC. When you have decrypted it, you will find that it is a person's name. Research the importance of this person's work during the Second World War.

SKILLS PROBLEM SOLVING
INTELLECTUAL INTEREST AND CURIOSITY
ADAPTABILITY
CO-OPERATION
RESPONSIBILITY

ACTIVITY

▼ ENCRYPTING A MESSAGE

1 The following is a message encrypted with a Caesar cipher. The shift is 16.
CEIJBO XQHCBUII
Decipher the message to give the original text.

2 Use a new Caesar cipher key to create an encoded message for a classmate. Do not include spaces between words, because this makes it harder to decrypt the message. Give them the encrypted message and see whether they can decrypt it and tell you the key.

A Caesar cipher is quite easy to crack, but most modern encryption is much more secure. There are two types of encryption:

- symmetric key encryption
- public key encryption.

SYMMETRIC KEY ENCRYPTION

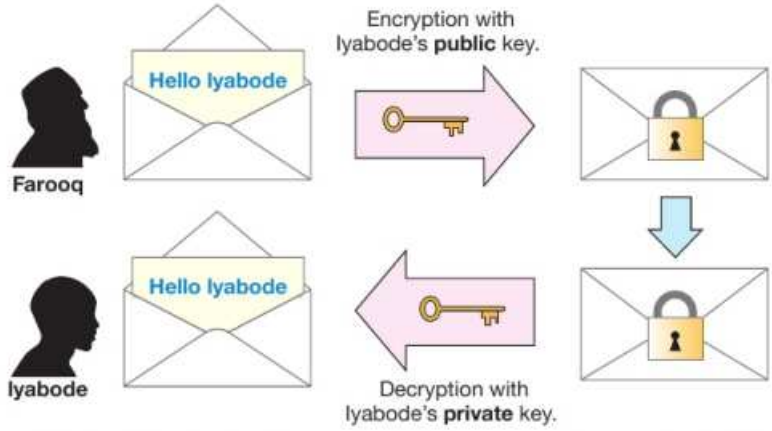
Symmetric key encryption uses the same key at both ends of the process, meaning that the same key is used to encrypt and decrypt the data.

PUBLIC KEY ENCRYPTION

Public key encryption uses two mathematically related keys called a key pair. One key is used to encrypt the data and a different key is used to decrypt it.

A computer shares a public key with other computers that want to send it encrypted data. This public key is mathematically related to a private key, which is not shared.

Figure 5.19 shows an example of public key encryption. If Farooq wants to send Iyabode an encrypted message, Farooq uses Iyabode's public key to encrypt the data. The data is then sent to Iyabode, who uses her private key to decrypt the message.



▲ **Figure 5.19** Farooq first sends Iyabode a request to send a secure message, then Iyabode shares her public key with Farooq

HINT

A key is not the same as a password.

Although Farooq's message won't be secure, because the public key is published, anyone who receives the message will not be able to understand it without Iyabode's private key.

SUBJECT VOCABULARY

eavesdropper an unauthorised person or piece of software that intercepts data from a private communication
packet a unit of data packaged to travel across a network

WIRELESS ENCRYPTION PROTOCOL (WEP)

It is easier to intercept data in a wireless network than in a wired network. Wireless Encryption Protocol (WEP) is used to secure the wireless transfer of data. It is the least secure wireless data encryption method. This is because every device on the wireless network uses the same key for every transfer. This means that if an **eavesdropper** studies enough **packets**, they can identify the key, and this provides them with unlimited access to all data from every device on the wireless network.

DID YOU KNOW?

Experts tested the strength of WEP by examining 10,000 packets in less than a minute. Having done this, they then took less than 3 seconds to identify the WEP encryption key used to encrypt the packets.

WI-FI PROTECTED ACCESS (WPA)

Wi-Fi Protected Access (WPA) is a security protocol designed to provide better encryption than WEP. WPA generates a new key for each device on the wireless network. New keys are also provided for each packet of data that is sent.

VIRTUAL PRIVATE NETWORK (VPN)

A VPN provides access to a private LAN from a remote location. The connection to the LAN is created using the infrastructure of a public network like the internet. Data sent using a VPN is encrypted so that it is secure if it is intercepted. An individual might use a VPN to:

- access their employer's network when working from home
- access computers in a different geographical location, perhaps to avoid the local restrictions on access to web content (such as due to censorship or **geolocation rights management**)
- make secure payments
- prevent surveillance of and access to their web activity.

SUBJECT VOCABULARY

geolocation rights management the use of a user's location to block access to services online; for example, a television channel might only allow users in their own country to watch their shows online

DID YOU KNOW?

Mobile, desktop and browser apps are available to make it easy for users to change their network settings so that they can use VPNs.

FILE ACCESS RIGHTS

File access rights are also known as file permissions. They can be set for individual files, folders or drives, and they ensure that users are either allowed to read only or allowed to read and write to the file, folder or drive.

TRANSACTION LOGS

SUBJECT VOCABULARY

log a record of events
transaction the exchange of data

All network activity can be recorded in a **log** file. Although this does not directly secure network data, a **transaction** log can help to identify which computers and network devices have been accessed. This can allow administrators to identify any unusual activity that might be a threat to data security.

BACKUPS

A backup is a copy of one or more files. The backup or backups are usually stored on a different storage device to the original file.

For more information about backups, see *Unit 1 Digital devices* (page 42) and *Unit 3 Operating online* (pages 104–105).

CHAPTER QUESTIONS

SKILLS PROBLEM SOLVING

1 Which **one** of these connects a LAN to a WAN? **(1)**

A Switch
B Gateway
C Modem
D Server

SKILLS DECISION MAKING

2 Explain why public key encryption is more secure than symmetric key encryption. **(2)**

SKILLS PROBLEM SOLVING

3 State which security device controls the traffic entering a network. **(1)**

SKILLS PROBLEM SOLVING

4 State **two** methods by which devices are identified to each other on a network. **(2)**

SKILLS REASONING

5 Explain why IPv6 was used to replace IPv4. **(2)**

SKILLS PROBLEM SOLVING

6 State **two** uses for a VPN. **(2)**

SKILLS REASONING PROBLEM SOLVING

7 State the reason why a booster is used in a network. **(1)**

SKILLS INTERPRETATION

8 Describe how encryption secures data. **(3)**

SKILLS PROBLEM SOLVING

9 List **three** methods of securing data on networks. **(3)**

SKILLS DECISION MAKING

10 State **three** benefits to users of using a local area network. **(3)**

SKILLS ADAPTIVE LEARNING EXECUTIVE FUNCTION ADAPTABILITY COMMUNICATION

11 Draw and label a network diagram for a home that includes:

- two smartphones
- one tablet device
- two desktop PCs
- internet access. **(8)**